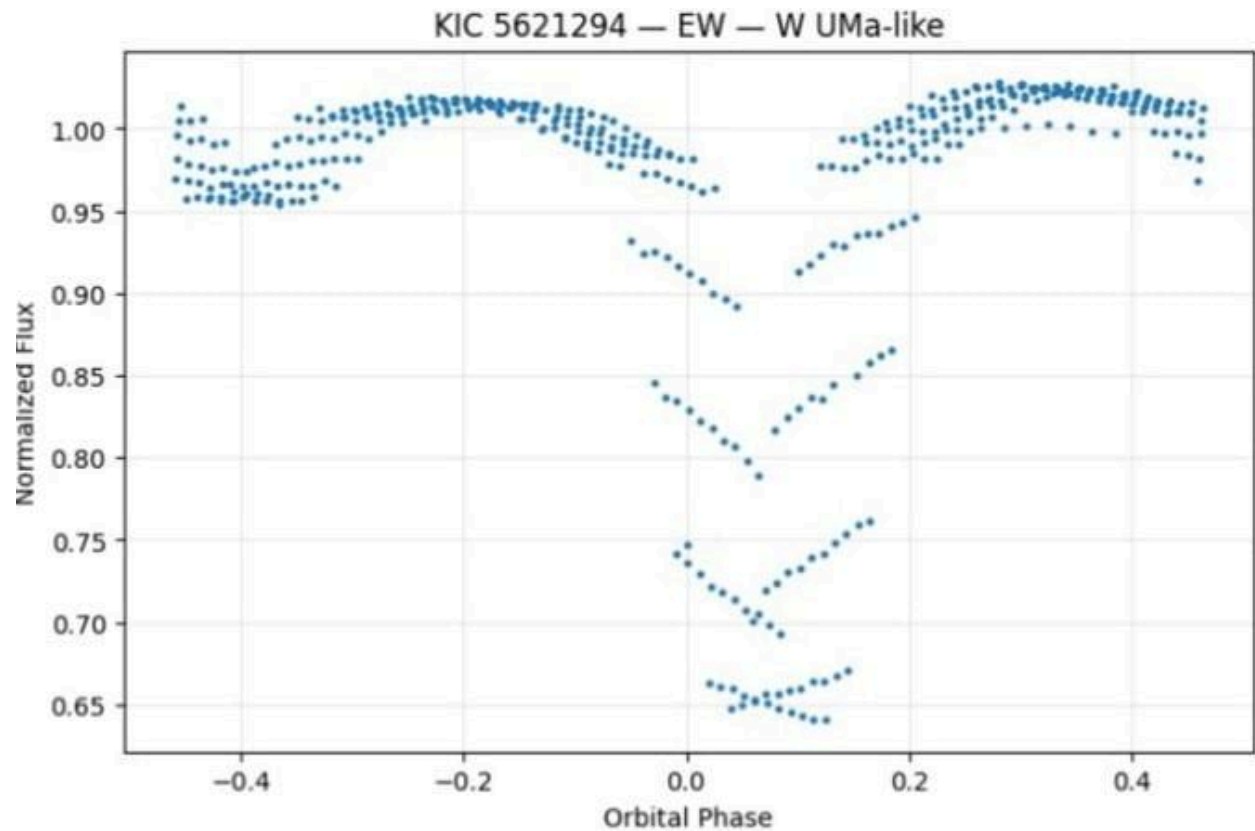


KIC 5621294

Outputs

Target: KIC 5621294
Mission: KEPLER
Period: 0.9295 days
Primary depth: 0.3593
Secondary depth: 0.2638
Depth ratio: 0.734

Program classification: EW



Interpretation:

1. KIC 5621294, the period observed and calculated are similar 0.928 days.
2. Failed the morphological classification due to just using the eclipse depth, which placed it in EW.

Potential solutions to this morphological issue:

1. We can create a separate morphological classification module which does the weighted sum of eclipse depth, eclipse width. For the constants we can use something along the lines of $D_i / \text{summation of all the } D_i$. Where D_i can be either the primary or secondary dip. Disadvantage of this is that we would lose information but doing so, but if the values of these constants have a physical motivation then it may work
2. A more precise decision making tree. If the depth ratio is closer to 0.7-0.885 which is the threshold for EB and EW, which is empirically found in my program we consider the period and other factors.